Daniel Lofeodo, M.Sc. Computer Science

AI/ML ENGINEER | SOFTWARE ENGINEER

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SUMMARY

Applied AI engineer with a background in professional machine learning and software engineering. Master's degree in machine learning from Yoshua Bengio's lab with 4.1 GPA. Experienced in working with stakeholders to translate requirements into shipped products. Experienced in creating AI-powered products, orchestrating agentic systems, building RAG systems, and evaluating and monitoring their cost and performance.

TECHNICAL SKILLS

Agentic AI & LLMs	Claude Code, LangGraph, MCP, tool calling, structured outputs, prompt engineering, guardrails
RAG & Retrieval	Embeddings, FAISS, pgvector, semantic search, reranking, citation grounding, retrieval evaluation
Programming & Cloud	Python, FastAPI, REST APIs, TypeScript, C#, C++, GCP, AWS, Azure, Docker, CI/CD, Databricks
Evals & Testing	Golden datasets, LLM-as-judge, automated tests in CI, prompt-injection testing, cost/drift monitoring
Machine Learning	PyTorch, scikit-learn, reinforcement learning, time series forecasting, Pandas, Spark
Languages	English (native), French (native), Italian (C1)

EXPERIENCE

Machine Learning Engineer (Master's Research Internship)

Apr 2025 - December 2025

Hydro-Québec, Montreal, QC · *Imitation learning of hydroelectric dam operation simulation under extreme flooding.*

- Trained, evaluated, and shipped multiple agents for hydro-dam operation capable of handling **1-in-10,000-year floods** consistently.
- Met weekly with stakeholders and translated their requirements into agent behaviour.
- Quantified trade-offs between state-of-the-art models, lightweight models, open-source models, and locally trained models.
- Built a realistic physics-based flood simulator within which agents were trained and evaluated.

Software Engineer

Aug 2021 – Jun 2024

Creaform, Montreal, QC · *C++/C#: engineering part inspection software for ultra-precise 3D scanners.*

- Co-built VXintegrity, a new inspection module earning **seven figures a year**.
- Collaborated weekly with stakeholders to materialize ideas from proof of concepts, to prototypes, to shipped products.
- Built a fully featured reverse engineering CAD module from the ground up in a team of 5 engineers.
- Owned the transition of 2 flagship products from C++/CLI/Winforms to C#/WPF, in a codebase of millions of lines.

SELECTED DEMOS

Latent SpaceMail: Autonomous Agentic AI Research Pipeline

May 2026 - Present

Built solo: fully automated weekly AI briefing run by 6 agents, live to subscribers 5 months, to stay current with emerging AI trends.

- 6 LLM agents working collaboratively to research, curate, and draft an automated newsletter for **0.90¢ per week**.
- Weekly evaluation of agent pipeline performance reported on a dashboard to catch model drift, check agent health, and monitor cost.
- Implemented and evaluated the cost and performance of a LangGraph agent with context management for deduplication.
- Built custom APIs using FastAPI for pipeline triggering and subscriptions.
- Used LLM-as-a-judge with structured output to rank news articles and scientific papers against a hand-written rubric.
- Quantitative comparative analysis of many models across different providers to select optimal models for different tasks.
- 100% automated on Google Cloud Platform using Cloud Scheduler and Pub/Sub to trigger and orchestrate agentic pipeline.
- Prompt injection guardrails implemented: successful attacks tested down **from 10/115 to 0/115**.
- Implemented automated tests in CI through GitHub Actions.
- Fully custom website with Google Authentication, issue browsing, settings personalization, Cloudflare DNS.

RAG With Receipts: Benchmarking RAG Accuracy, Grounding, and Latency

Sep 2026 - Oct 2026

Built-solo: RuneScape wiki RAG system, hosted on Google Cloud Platform, designed to optimize latency backed by evaluation metrics

- Different rerankers tested for performance and latency: found small reranker to be **4x faster and more accurate**.
- Performance tested against hand-labeled golden dataset: **96% right source**, 84% correct, 5 of 5 off-topic questions declined.
- Every answer cites its source, refusing to answer if no source is found in the corpus.
- Models tested for cost, accuracy, and performance: cheaper and faster model shows no statistically significant performance difference.

EDUCATION & CERTIFICATIONS

M.Sc. Computer Science (Machine Learning) · Mila, University of Montreal · *GPA 4.1/4.3* — Aug 2024 - May 2026

B.Eng. Mechanical Engineering · Concordia University · *Dean's List (2021)* — Aug 2017 - May 2021

Certifications	AWS Certified Machine Learning Engineer - Associate (MLA) · Claude Code in Action (Anthropic)
Conference	Autonomous Hydroelectric Dam Operation under Extreme Flooding · Mila Annual Conference